



Manual for RiPEST

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1 INTRODUCTION

This tool is developed as an interactive guide to help the user to plan and execute a statistically sound and risk-based survey on plant pests. It has been developed based on the following EFSA guidance document:

- EFSA (European Food Safety Authority), Lázaro E, Parnell S, Vincent Civera A, Schans J, Schenk M, Cortiñas Abrahantes J, Zancanaro G and Vos S, 2020a. General guidelines for statistically sound and risk-based surveys of plant pests. EFSA supporting publication 2020:EN-1919. 65 pp. [doi:10.2903/sp.efsa.2020.EN-1919](https://doi.org/10.2903/sp.efsa.2020.EN-1919)

The online application RiBESS+ (Risk based surveillance systems) is used for the underlying calculations. It implements statistical methods for estimating the sample size, design prevalence (achieved design prevalence), global (and group) sensitivity (achieved confidence level) and probability of freedom from disease.

Tip: A session can be downloaded at any point during the survey process. This way the survey can be continued or edited at a later point in time.

Note: No questions can be skipped, unless it is specifically indicated as optional.

The user can start a new survey by clicking on 'Start New Survey' on the welcome page, then the user will be redirected to the 'Choose your survey'-page.

Note: Switching between surveys is not possible. If the user wants to start a new survey, then the application should be restarted by pressing the 'house'-icon in the top right corner of the tool.

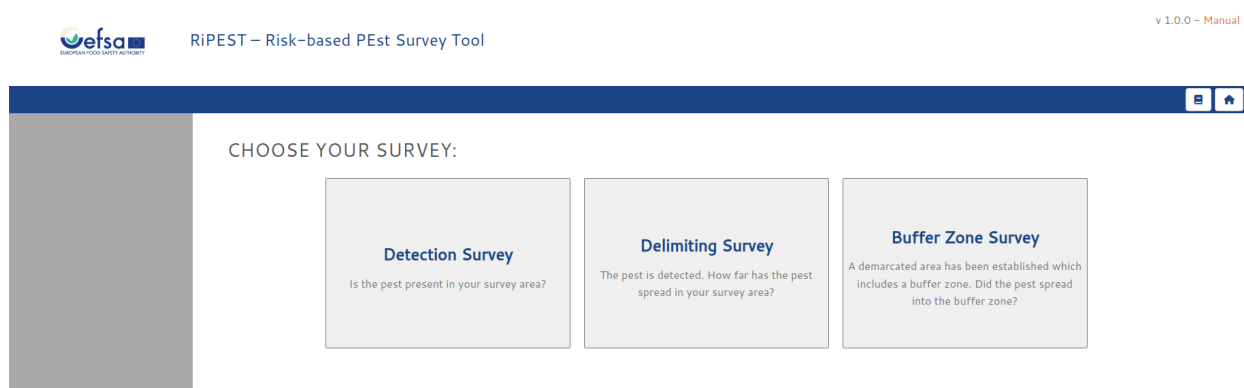


Figure 1: Choose your survey.

2 DETECTION SURVEY

The detection survey conducts a survey to assess if the pest is present in the survey area.

The detection survey is divided into five sections: Initiation, Preparation, Design, Implementation and Conclusion.

The figures included in this section show a made up survey as an example.

2.1 Initiation phase

The initiation phase is divided into three subsections: 1. Motivation, 2. The pest and 3. Survey area & host plants. The goal is to collect some general information about these topics.

Every section starts with a short introduction.

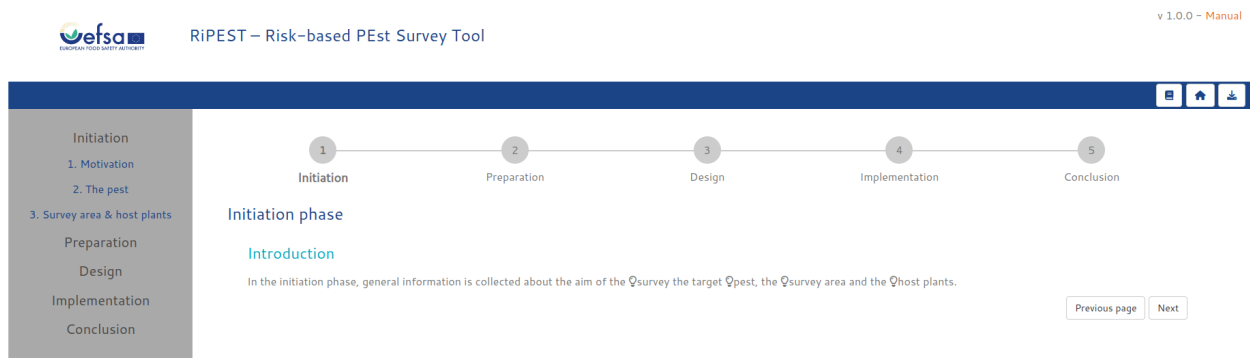


Figure 2: The first page of the detection survey.

The user is able to navigate through the survey using the menu at the left side of the page (the gray bar) or using the buttons 'Previous page' and 'Next' at the bottom right of the page.

At any point during the survey the user can consult the glossary, pressing a button in the top right of the page. If a word from the glossary appears for the first time in the tool, it will be highlighted by the lightbulb icon. When hovering over the word, a tooltip will show containing the definition of the word as can be found in the glossary.

In the top right of the page, the user can always download a session manually. Or the user can restart the tool by clicking on the 'house'-button, this should be done carefully since already entered data will be lost if the session was not downloaded.

Clicking on 'Next' will bring the user to the first subsection of the initiation phase, the motivation.

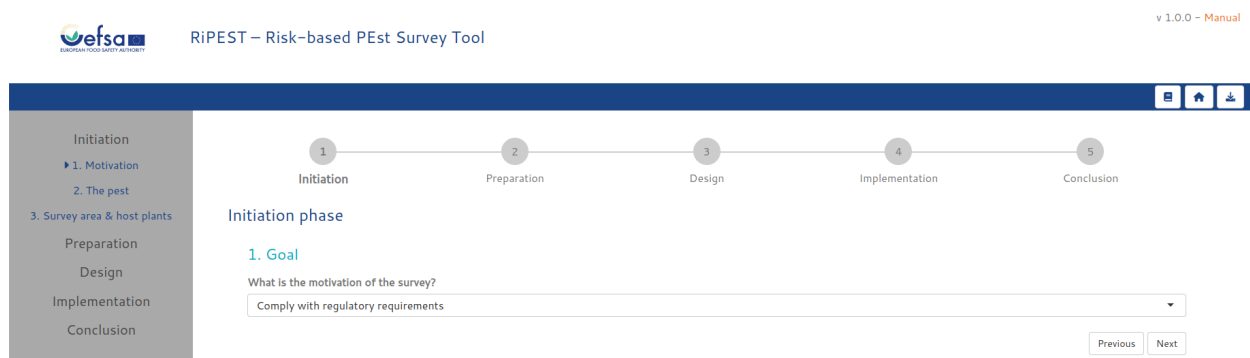


Figure 3: Motivation of the detection survey.

Next, the user will have to answer some details about the pest.

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RiPEST – Risk-based PEst Survey Tool

v 1.0.0 – Manual

Initiation
1. Motivation
▶ 2. The pest
3. Survey area & host plants
Preparation
Design
Implementation
Conclusion

1 Initiation 2 Preparation 3 Design 4 Implementation 5 Conclusion

Initiation phase

2. The pest

In case the survey is aimed to detect multiple pests at the same time, this procedure needs to be done multiple times. This tool only considers one pest at a time.

What is the target pest of the survey?

Acidiella kagoshimensis

What is the status of the pest in your target population?

Absent

When was the last survey conducted for the same pest and target population?

Select a year

Please provide the year of the last survey.

2020

Which survey(s) was(were) conducted in that year?

Detection survey

Previous Next

Figure 4: Details of the pest.

The tool is connected to a database containing all known and relevant information for a certain pest. All pests in the dropdown menu are included in this database and some information relevant for that pest will automatically be extracted. In case the user wants to survey a pest that is not included in the list, the user can just type in the pest and press enter.

Disclaimer: The database is in the process of being populated, therefore some information might be missing.

Continuing to the next page.

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RiPEST – Risk-based PEst Survey Tool

v 1.0.0 – Manual

Initiation
1. Motivation
2. The pest
▶ 3. Survey area & host plants
Preparation
Design
Implementation
Conclusion

1 Initiation 2 Preparation 3 Design 4 Implementation 5 Conclusion

Initiation phase

3. Survey area & host plants

Member State

Spain

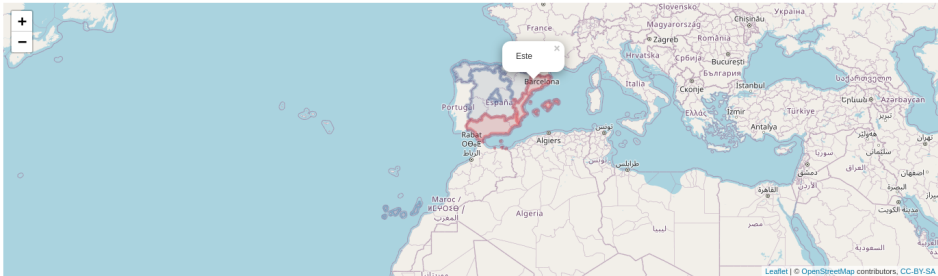
What is the survey area?

Select the area (NUTS 0,1,2,3)

Select the appropriate NUTS classification.

NUTS 1

Selected NUTS: Sur, Este



Leuven | © OpenStreetMap contributors, CC-BY-SA

Figure 5: Description of the survey area and host plants.

The default option for describing the survey area is done by using the NUTS (Nomenclature of territorial units for statistics) classification. If this option is used, a map will appear and the user will have to select the relevant NUTS on the map (selected regions will appear in red, see Figure 5).

2.2 Preparation phase

The preparation phase will handle more details about the actual survey. After the short introduction, this phase is also divided into three subsections: 1. Characterise the pest, 2. Target population and 3. Inspection units.

Note: Uploading shapefiles is always optional and are only used for reporting.

The screenshot shows the 'Detection survey' interface of the RiPEST tool. A progress bar at the top indicates five steps: 1. Initiation, 2. Preparation, 3. Design, 4. Implementation, and 5. Conclusion. The 'Preparation' step is currently active. On the left, a sidebar lists the steps: Initiation, Preparation, 1. Characterise the pest (selected), 2. Target population, 3. Inspection units, Design, Implementation, and Conclusion. The main content area for 'Preparation phase' shows '1. Characterise the pest'. Below this, it says 'Upload a shapefile of the environment's suitability for the targeted pest. (Optional)' and provides a 'Browse...' button with the text 'No file selected'. At the bottom right, there are 'Previous' and 'Next' buttons.

Figure 6: Characteristics of the pest.

Continuing to the next page, the user will have to provide the inspections unit(s).

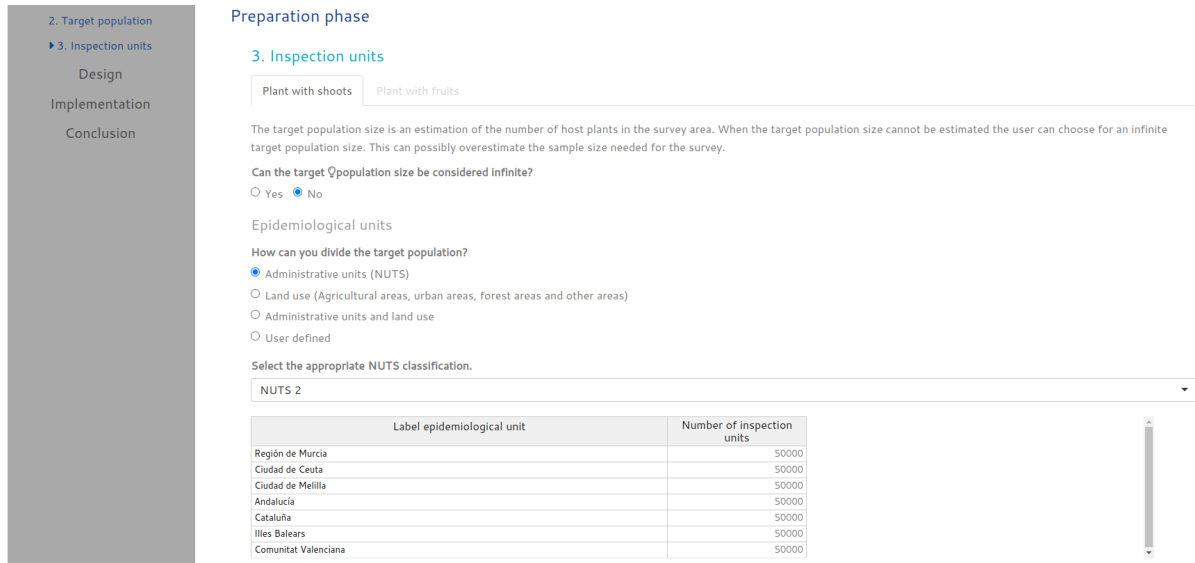
The screenshot shows the 'Detection survey' interface of the RiPEST tool, continuing from the previous step. The progress bar and sidebar are the same. The main content area for 'Preparation phase' now shows '2. Target population'. It includes the text: 'The target population can be defined through the inspection units, the epidemiological units and possible risk factors that could increase the probability of infection of the host plants. Please note that you cannot combine plants, traps and/or environmental samples into a single survey design.' Below this, it asks 'What is(are) the inspection unit(s) considered in the survey?' and provides a text input field containing 'Plant with shoots' and 'Plant with fruits'. At the bottom right, there are 'Previous' and 'Next' buttons.

Figure 7: Description of the survey area and host plants.

The user has to select at least one inspection unit. If multiple inspection units are selected, the user will have to carry out a separate detection survey for each inspection unit. In this case, two inspection units are selected: Plant with shoots and Plant with fruits.

In the following section, the first set of questions deals with the size of the target population divided into epidemiological units. The tool provides three predefined options: only using NUTS, using land

use and using a combination of both. The user has also the option to define their own epidemiological units. For every epidemiological unit the number of inspections units should be given.



Preparation phase

3. Inspection units

Plant with shoots | Plant with fruits

The target population size is an estimation of the number of host plants in the survey area. When the target population size cannot be estimated the user can choose for an infinite target population size. This can possibly overestimate the sample size needed for the survey.

Can the target population size be considered infinite?

☐ Yes ☒ No

Epidemiological units

How can you divide the target population?

☒ Administrative units (NUTS)

☐ Land use (Agricultural areas, urban areas, forest areas and other areas)

☐ Administrative units and land use

☐ User defined

Select the appropriate NUTS classification.

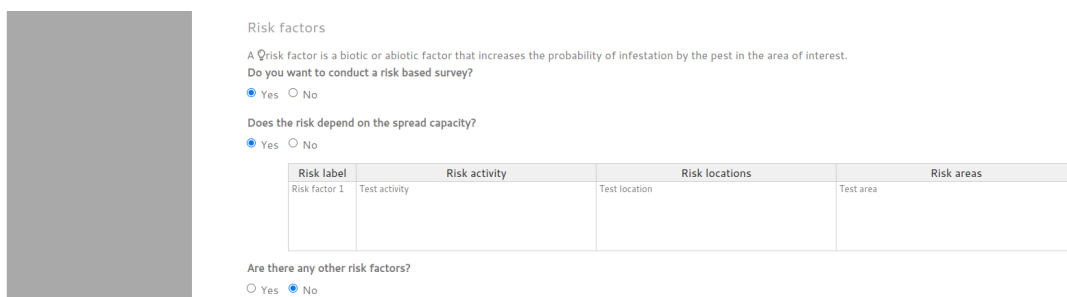
NUTS 2

Label epidemiological unit	Number of inspection units
Región de Murcia	50000
Ciudad de Ceuta	50000
Ciudad de Melilla	50000
Andalucía	50000
Cataluña	50000
Illes Balears	50000
Comunitat Valenciana	50000

Figure 8: Target population.

Notice that there are two tabs for subsection '3. Inspections units'. Both tabs contain the same set of questions, that must be filled in to be able to continue the survey.

The second set of questions deals with the risk factors. At most two risk factors can be considered per inspection unit.



Risk factors

A risk factor is a biotic or abiotic factor that increases the probability of infestation by the pest in the area of interest.

Do you want to conduct a risk based survey?

☒ Yes ☐ No

Does the risk depend on the spread capacity?

☒ Yes ☐ No

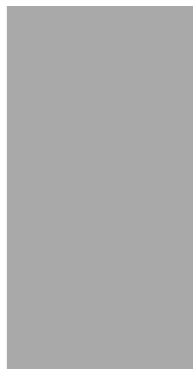
Risk label	Risk activity	Risk locations	Risk areas
Risk factor 1	Test activity	Test location	Test area

Are there any other risk factors?

☐ Yes ☒ No

Figure 9: Description of the risk factors.

The last set of questions deals with the detection and identification methods. It is possible to use two detection methods in parallel, in that case two sets of the same questions will have to be answered in two tabs.



Characterization of survey sites

The survey sites can be characterized according to the type of locality. Select the most appropriate description and select multiple options in case different types of localities are among the survey sites. You can also define your own survey site, by typing a description and adding it to the drop down list.

What is the type of survey site?

1.1 Field (arable, pasture) 1.2 Orchard/vineyard 1.3 Nursery 1.4 Forest

Survey period start: 08-2023

Survey period end: 12-2023

Detection & identification methods

A detection and identification method implies a process that usually involves a sequence of procedures. The sensitivity of the entire process (i.e. detection & identification method) must be determined. This so-called method sensitivity is defined as the probability that a truly positive inspection unit will be confirmed as positive.

You will be asked to characterize the detection & identification method used to survey the target pest by defining the procedures used in the field and in the lab. You may also have multiple detection and identification methods to survey a single pest.

How many detection methods will you use? 1

Figure 10: Detection and identification methods (Part 1).



Method 1

What type of survey procedure is applied in the field to detect the pest?

Laboratory testing

Which sampling matrix or matrices are collected?

Branch Crown

The sensitivity of the field procedure is termed sampling effectiveness, which is defined as the probability to collect an infested sample from an infested unit.

What is the ϕ sampling effectiveness? [%]

90

The sensitivity of the laboratory procedure is characterized by the diagnostic sensitivity, which is defined as the probability to detect the pest in an infested sample, following a specific sampling and/or diagnostic protocol. Such protocol may involve multiple steps (e.g. screening, detection and confirmation). These steps are usually sequential, meaning that samples only proceed to the next step when positive (e.g. confirmation is only performed when detection is positive). The sensitivity of each step should be known or should be estimated in order to calculate the diagnostic sensitivity of the laboratory procedure.

Please describe the main steps (e.g. screening, isolation, extraction, sample preparation, detection, confirmation) in the laboratory procedure:

Test steps

Please select the specific laboratory methods.

DAS-ELISA

What is the ϕ diagnostic sensitivity of the full laboratory procedure? [%]

95

Figure 11: Detection and identification methods (Part 2).

2.3 Design phase

The design phase deals with calculating the sample size per epidemiological unit and per risk location.

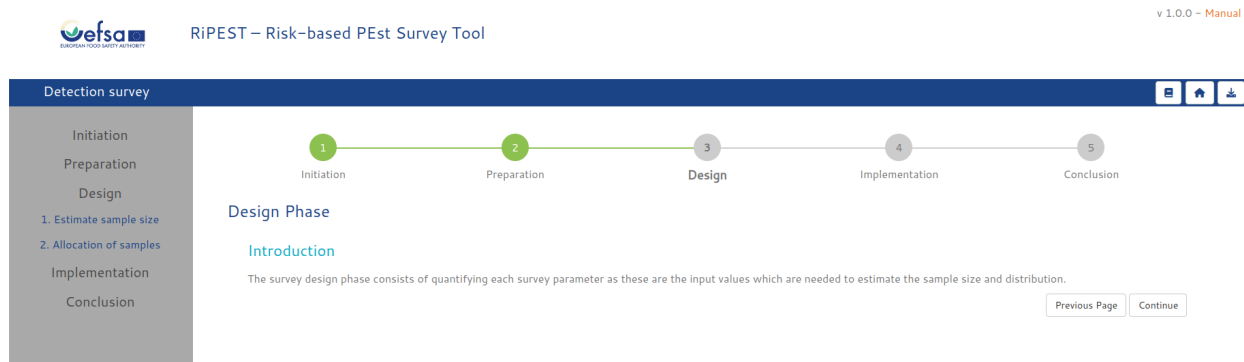


Figure 12: Introduction page of the design phase.

The design phase is divided into two subsections: 1. Estimate sample size and 2. Allocation of samples.

The overall confidence level of the survey is the first question of the design phase with a default of 95%. Be aware that this covers all inspection units, if there are multiple.

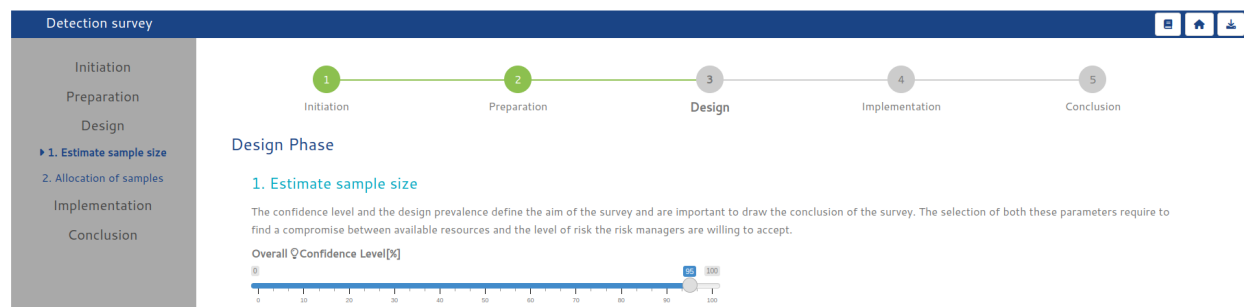


Figure 13: The overall confidence level of the survey.

The next questions are inspection unit specific. Therefore they need to be filled in for all inspection units. First, the design prevalence can be altered the default is set on 1%. Then, more details about the risk factors need to be provided and also the user can indicate if convenience sampling should be applied.

Design prevalence

Design prevalence [%]

1

Risk factor 1

Risk factors

How many risk levels do you want to take into account?

2

Epidemiological unit	Risk level	Relative risk	Percentage within each epidemiological unit [%]
Región de Murcia	high	2.00	5.00
Región de Murcia	base	1.00	95.00
Ciudad de Ceuta	high	2.00	5.00
Ciudad de Ceuta	base	1.00	95.00
Ciudad de Melilla	high	2.00	5.00
Ciudad de Melilla	base	1.00	95.00
Andalucía	high	2.00	5.00
Andalucía	base	1.00	95.00
Cataluña	high	2.00	5.00
Cataluña	base	1.00	95.00
Illes Balears	high	2.00	5.00
Illes Balears	base	1.00	95.00
Comunitat Valenciana	high	2.00	5.00
Comunitat Valenciana	base	1.00	95.00

Convenience sampling

Convenience sampling approach

No convenience sampling

Simulate

Figure 14: Design prevalence, risk factors and convenience sampling.

If everything is filled properly, then the user can press 'Simulate' and a table will show with the theoretical samples sizes in order to achieve the selected confidence level with the selected design prevalence.

Sample Size Table

The overall achieved confidence level: 80.06%

Epidemiological unit	Risk level	Population size	Theoretical sample size	Theoretical achieved confidence level
Región de Murcia	high	2500.00	7.00	0.11
Región de Murcia	base	47500.00	14.00	0.11
Ciudad de Ceuta	high	2500.00	7.00	0.11
Ciudad de Ceuta	base	47500.00	14.00	0.11
Ciudad de Melilla	high	2500.00	7.00	0.11
Ciudad de Melilla	base	47500.00	14.00	0.11
Andalucía	high	2500.00	7.00	0.11
Andalucía	base	47500.00	14.00	0.11
Cataluña	high	2500.00	7.00	0.11
Cataluña	base	47500.00	14.00	0.11
Illes Balears	high	2500.00	7.00	0.11
Illes Balears	base	47500.00	14.00	0.11
Comunitat Valenciana	high	2500.00	7.00	0.11
Comunitat Valenciana	base	47500.00	14.00	0.11

The confidence level for each risk group, i , is calculated ensuring that they have the same achieved confidence and it is based on the solution of the following formula $OC_L = 1 - \prod_{i=1}^K (1 - CLRG_i) = 1 - (1 - CLRG)^K$, where OC_L is the desired overall confidence level, $CLRG_i$ is the desired confidence level for each risk group, which in the case that is assumed to be the same for all risk group is referred as $CLRG$ without subscript and K represents the total number of risk groups identified.

Previous Continue

Figure 15: The sample size table.

Once the sample sizes have been estimated, the user can re-allocate the sample sizes. This step is optional. Allocation can be done proportional to the number of risk locations (only available if the survey is risk based) or proportional to the host plant population (always available). In the example survey in Figure 17, the samples have been allocated proportional to the number of risk locations. Keep in mind that the proportions in this case are grouped by risk levels.

Design

1. Estimate sample size

➔ 2. Allocation of samples

Implementation

Conclusion

Design Phase

2. Allocation of samples

Once the overall sample size has been calculated, the sample size within each epidemiological unit and/or risk level needs to be determined. The sample size can e.g. be the same in all epidemiological units or can be proportional to the number of inspection units in that unit.

Plant with shoots Plant with fruits

Do you want to reallocate the samples to the epidemiological units?

☒ Yes ☐ No

How do you want to allocate the samples?

Proportional to the number of risk locations

Only the column 'Number of risk locations' is editable for risk levels medium and/or high.

Risk level	Population size	Theoretical sample size	Epidemiological unit	No. of risk locations	Proportion of risk locations	Updated theoretical sample size
high	2500	7	Región de Murcia	10	0.0625	4
high	2500	7	Ciudad de Ceuta	20	0.125	7
high	2500	7	Ciudad de Melilla	30	0.1875	10
high	2500	7	Andalucía	40	0.25	13
high	2500	7	Cataluña	10	0.0625	4
high	2500	7	Illes Balears	20	0.125	7
high	2500	7	Comunitat Valenciana	30	0.1875	10
base	47500	14	Región de Murcia	0	0	14
base	47500	14	Ciudad de Ceuta	0	0	14
base	47500	14	Ciudad de Melilla	0	0	14
base	47500	14	Andalucía	0	0	14
base	47500	14	Cataluña	0	0	14
base	47500	14	Illes Balears	0	0	14
base	47500	14	Comunitat Valenciana	0	0	14

Previous Continue

Figure 16: Sample size allocation proportional to the number of risk locations.

2.4 Implementation phase

The implementation phase helps to select the specific survey sites within a survey area at random. The user can choose to include the geographic coordinates. Then the user can download a template and provide all known inspection units for a specific epidemiological unit (combined with risk factors if relevant). After uploading a file containing all relevant inspection units, a new button will become available (See Figure 17). The tool will pick the inspection units to be sampled in the field at random to ensure the validity of the survey.

Detection survey

Initiation

Preparation

Design

Implementation

Conclusion

1 2 3 4 5
 Initiation Preparation Design Implementation Conclusion

Implementation Phase

In the implementation phase, the user selects specific survey sites within the survey area.

At this stage, the survey area is divided into epidemiological units. For each epidemiological unit, the tool has calculated the needed sample size to reach the desired overall confidence level. In this phase, the survey sites need to be selected within each epidemiological unit and/or risk location. The user has to provide a list with inspection units IDs and optionally the geographic coordinates of the inspection units. The user can download a template in excel to provide the tool with the data in the correct format. Note that the sheet names are the epidemiological units plus the risk levels in case there are any risk factors.

Plant with shoots Plant with fruits

Do you want to include the geographic coordinates of the inspection units?

☒ Yes ☐ No

Step 1: Download the template.

Download template

Step 2: Complete the list of inspection units.

Step 3: Upload the template again.

Upload template

Browse... Plant with shoots_samples.xls

Upload complete

Download inspection units to sample

Previous Page Finish survey

Figure 17: Implementation phase.

At this point the theoretical part is done and the user has to continue with the field work in order to come to a conclusion in the next step.

2.5 Conclusion

In the conclusion, the user will be able to report the findings and to download two types of reports. One report will be in MS Word and the other in MS Excel (EUROPHYT). The report in MS Word will be for every inspection unit separately. The tool for EUROPHYT will include all inspection units.

In case the pest is not found the user will be asked to provide the number of negative samples and the number of undetermined samples. Only if these values are provided, the tool will be able to provide the user with a conclusion.

Detection survey

Initiation Preparation Design Implementation Conclusion

Conclusion

Was the pest found during the survey?

☐ Yes ☒ No

Plant with shoots Plant with fruits

Refresh/update

	Risk level	Population size	Theoretical sample size	Epidemiological unit	No. of risk locations	Proportion of risk locations	Updated theoretical sample size	Negative samples
1	high	2500	7	Región de Murcia	10	0.0625	4	4
2	high	2500	7	Ciudad de Ceuta	20	0.125	7	7
3	high	2500	7	Ciudad de Melilla	30	0.1875	10	10
4	high	2500	7	Andalucía	40	0.25	13	13
5	high	2500	7	Cataluña	10	0.0625	4	4
6	high	2500	7	Illes Balears	20	0.125	7	7
7	high	2500	7	Comunitat Valenciana	30	0.1875	10	10
8	base	47500	14	Región de Murcia	0	0	14	14
9	base	47500	14	Ciudad de Ceuta	0	0	14	14
10	base	47500	14	Ciudad de Melilla	0	0	14	14
11	base	47500	14	Andalucía	0	0	14	14
12	base	47500	14	Cataluña	0	0	14	14
13	base	47500	14	Illes Balears	0	0	14	14
14	base	47500	14	Comunitat Valenciana	0	0	14	14

Calculate achieved confidence level

Figure 18: The findings of the survey (user input).

Risk level	Population size	Theoretical sample size	Epidemiological unit	No. of risk locations	Proportion of risk locations	Updated theoretical sample size	Negative samples	Undetermined samples	Actual achieved confidence level
high	2500	7	Región de Murcia	10	0.0625	4	4	0	0.0641
base	47500	14	Región de Murcia	0	0	14	14	0	0.1081
high	2500	7	Ciudad de Ceuta	20	0.125	7	7	0	0.1095
base	47500	14	Ciudad de Ceuta	0	0	14	14	0	0.1081
high	2500	7	Ciudad de Melilla	30	0.1875	10	10	0	0.1528
base	47500	14	Ciudad de Melilla	0	0	14	14	0	0.1081
high	2500	7	Andalucía	40	0.25	13	13	0	0.194
base	47500	14	Andalucía	0	0	14	14	0	0.1081
high	2500	7	Cataluña	10	0.0625	4	4	0	0.0641
base	47500	14	Cataluña	0	0	14	14	0	0.1081

Conclusion for inspection unit Plant with shoots: The survey area is free from the pest, based on a survey with confidence level of 81.95% and a design prevalence of 1%.

Figure 19: The actual conclusion for the relevant inspection unit.

If there are multiple inspection units, the tool will also provide an overall conclusion.

In case the pest is found, the user is still asked to provide the number of positive samples, the number of negative samples and the number of undetermined samples (See Figure 20).

Conclusion

Was the pest found during the survey?
☒ Yes ☐ No

Plant with shoots Plant with fruits

	Risk level	Population size	Theoretical sample size	Epidemiological unit	No. of risk locations	Proportion of risk locations	Updated theoretical sample size	Positive samples	Negative samples	Undetermined samples
1	high	2500	7	Región de Murcia	10	0.0625	4			
2	high	2500	7	Ciudad de Ceuta	20	0.125	7			
3	high	2500	7	Ciudad de Melilla	30	0.1875	10			
4	high	2500	7	Andalucía	40	0.25	13			
5	high	2500	7	Cataluña	10	0.0625	4			
6	high	2500	7	Illes Balears	20	0.125	7			
7	high	2500	7	Comunitat Valenciana	30	0.1875	10			
8	base	47500	14	Región de Murcia	0	0	14			
9	base	47500	14	Ciudad de Ceuta	0	0	14			
10	base	47500	14	Ciudad de Melilla	0	0	14			
11	base	47500	14	Andalucía	0	0	14			
12	base	47500	14	Cataluña	0	0	14			
13	base	47500	14	Illes Balears	0	0	14			
14	base	47500	14	Comunitat Valenciana	0	0	14			

Download report

Overall conclusion: The pest was found! Please conduct a delimiting survey accordingly.

Download report EUROPHYT

Previous Page

Figure 20: The findings of the survey in case the pest is found.

3 DELIMITING SURVEY

The delimiting survey helps the user to demarcate an area around a single finding or a group of findings. The user will be guided through the survey using a made up example survey.

The structure of the delimiting survey is similar to the detection survey. It starts with an initiation phase, followed by a preparation phase, a design phase and a conclusion.

3.1 Initiation Phase

In the initiation phase, general information is collected about the aim of the survey, the target pest, the survey area and the host plants. It is divided into three subsections: 1. Motivation, 2. The pest and 3. Survey area & host plants.

The motivation of the survey has been prefilled, thus the user can continue to the subsection about the pest. Here the user will have to provide specific information about the findings. Keep in mind that the label(s) of the finding(s) will be used later on, in the survey, in an interactive map.

The screenshot shows the 'Initiation phase' of a survey preparation process. The left sidebar lists the steps: Initiation (1. Motivation, 2. The pest, 3. Survey area & host plants), Preparation, Design, and Conclusion. The main content area is titled 'Initiation phase' and '2. The pest'. It contains several form fields and a table.

Form fields:

- What is the target pest of the survey? (Dropdown menu with 'Acidiella kagoshimensis' selected)
- What was the status of the pest in the outbreak area prior to the finding? (Dropdown menu with 'Absent' selected)
- When was the last detection survey conducted without finding the pest for the same target population? (Dropdown menu with 'Select a year' selected)
- Please provide the year of the last survey. (Text input with '2021' entered)
- How many findings have been made for the pest? (Text input with '3' entered)

Table:

	Label finding	Latitude	Longitude	Date	Field detection method
1	Finding 1	38.678743	-0.287047	08/01/2023	Visual examination (incl. macroscopic identification of the pest and sampling for laboratory analysis)
2	Finding 2	38.704183	-0.199419	08/02/2023	Visual examination (incl. macroscopic identification of the pest and sampling for laboratory analysis)
3	Finding 3	38.991888	-0.288743	08/03/2023	Visual examination (incl. macroscopic identification of the pest and sampling for laboratory analysis)

Buttons: Previous, Next

Figure 21: The pest.

Continuing to the next page.

The screenshot shows the 'Initiation phase' of a survey preparation process. The left sidebar lists the steps: Initiation (1. Motivation, 2. The pest, 3. Survey area & host plants), Preparation, Design, and Conclusion. The main content area is titled 'Initiation phase' and '3. Survey area & host plants'. It contains several form fields and a note.

Form fields:

- Member State (Dropdown menu with 'Spain' selected)
- Which host plants or environmental components (soil, water...) are targeted by the survey? (Text input with 'Other' entered)
- Please describe what is targeted by the survey. (Text input with 'Test host' entered)
- Do you have a shapefile with the host plant distribution? (Radio buttons: Yes, No. 'No' is selected)

Note: Shapefiles are only used in the reporting.

Buttons: Previous, Continue

Figure 22: Survey area and host plants.

3.2 Preparation phase

The main goal of the preparation phase is to construct the potential infested zone around each finding or group of findings. In order to do this, this phase is divided into three subsections: 1. Source(s) of infestation, 2. Estimate boundaries and 3. Delimit boundaries.

First, the user will have to indicate the annual short-distance spread rate of the pest in meters. Based on this value, the radius of the potential infested zone will be calculated.

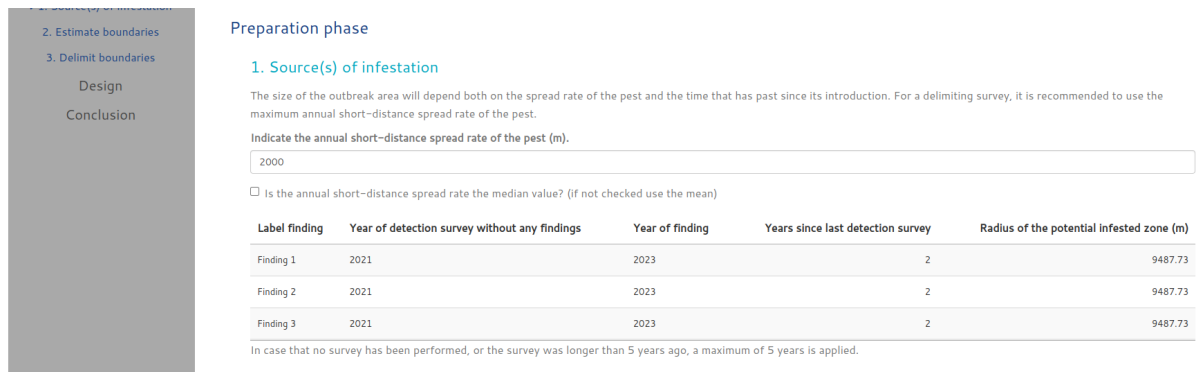


Figure 23: Source(s) of infestation.

Next, the user has the option to add potential risk locations by uploading an MS Excel file. The orange markers on the map represent these locations (See Figure 24). Hovering over the map will show the relevant label of the finding or the risk location.

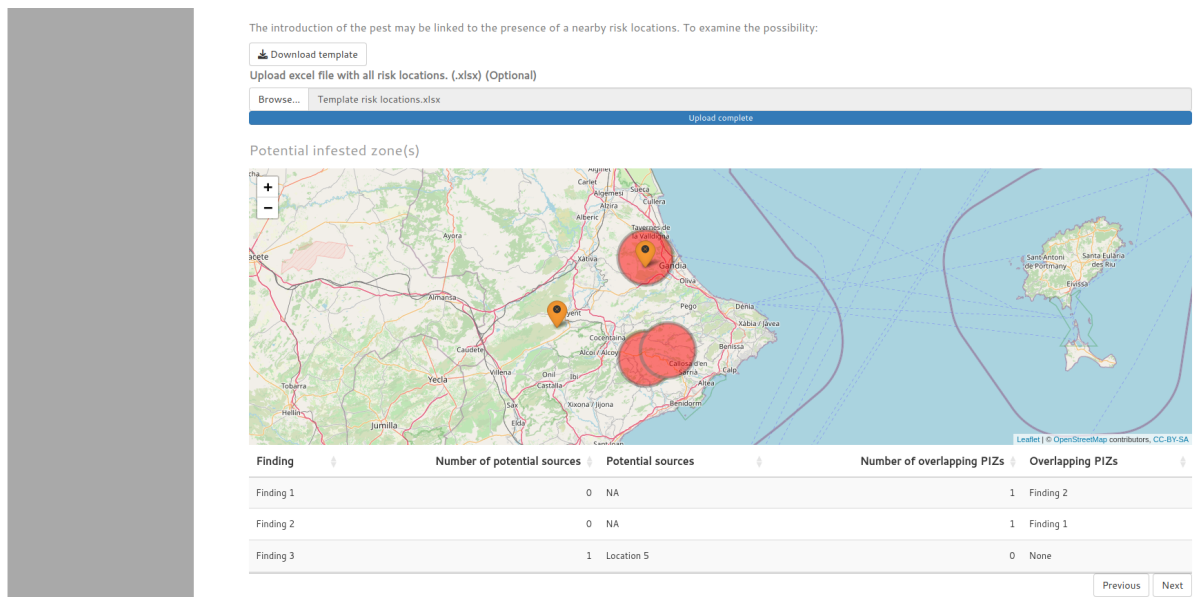


Figure 24: Potential infested zone(s) and risk locations.

Beneath the map, a summary table is shown. It contains the information shown on the map.

In the following subsection, the independent findings are split into separate delimiting surveys as can be seen in Figure 25. From now on, every page will have two tabs: Delimiting survey 1 and Delimiting survey 2.

Tip: If the findings are independent a separate session could be run with the tool. There is no need to combine these surveys into one session. The tool just helps to establish if the findings can indeed be seen as independent.

The screenshot shows the 'Delimiting survey' web application interface. On the left is a sidebar with a navigation menu containing: Initiation, Preparation (selected), 1. Source(s) of infestation, 2. Estimate boundaries (active), 3. Delimit boundaries, Design, and Conclusion. The main content area is titled 'Preparation phase' and shows a progress bar with five steps: 1. Initiation, 2. Preparation (current), 3. Design, 4. Implementation, and 5. Conclusion. Below the progress bar, the section '2. Estimate boundaries' is displayed. It contains a text block explaining that in the case of overlapping findings, there are two approaches: the recommended approach (combining areas) and the conservative approach (building a new circular zone around the centroid). Below this text are two tabs: 'Delimiting survey 1' (selected) and 'Delimiting survey 2'. Under 'Delimiting survey 1', it shows 'Overlapping findings: Finding 1, Finding 2' and 'Potential Source(s): None'. It then asks to 'Choose the approach for performing the delimiting survey.' with two radio buttons: 'Recommended approach' (selected) and 'Conservative approach'. At the bottom right are 'Previous' and 'Next' buttons.

Figure 25: Estimate boundaries.

In the case of overlapping findings (meaning that the potentially infested zones overlap) there are two approaches for conducting the delimiting survey. The area to be surveyed can be defined by combining the areas covered by the initially defined potentially infested zones (the recommended approach) or by building a new circular potentially infested zone around the centroid of the findings which encompasses all initially defined potentially infested zones (the conservative approach).

In case of only one finding in the neighbourhood of a risk location, the risk location will be taken as new center of the PIZ.

In the last subsection, the band to be inspected is constructed (the green band in the Figures). In Figure 26 and Figure 27, the recommended approach and the conservative approach are shown.

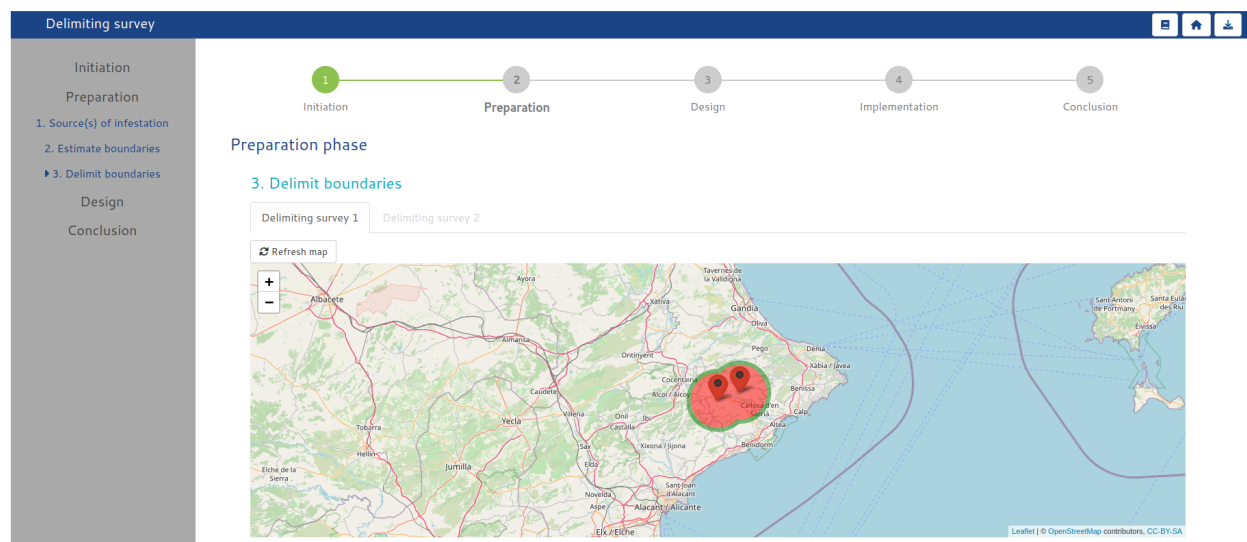


Figure 26: The recommended approach.

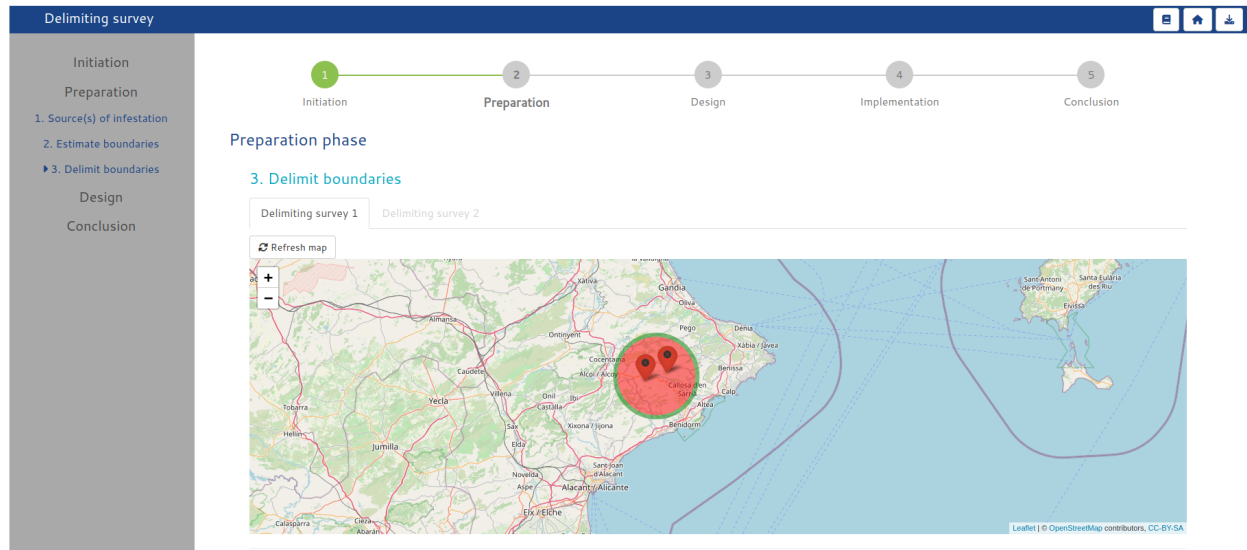


Figure 27: The conservative approach.

As a last step of the preparation phase, the user will have to provide information about the detection and identification methods. This is set of questions are completely the same as for the detection survey.

3.3 Design phase

The design phase will help the user to conduct a survey within the green band constructed in the preparation phase. Depending on the findings in this survey, the PIZ will increase or decrease.

First, the user has to set the confidence level and the design prevalence, which have a default value of 95% and 1% respectively.

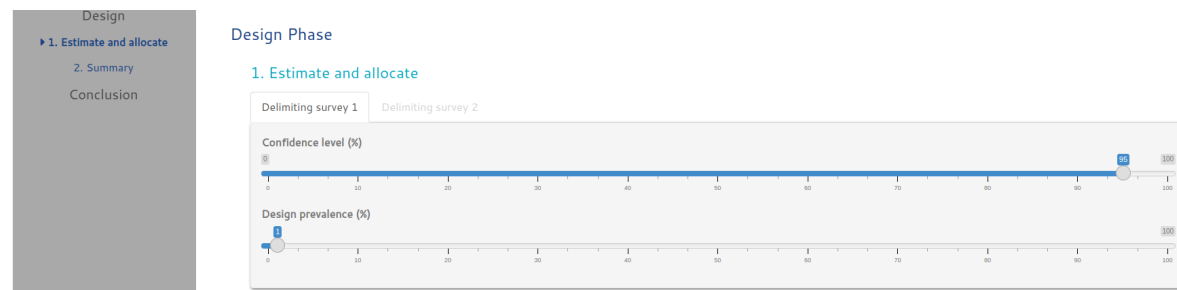


Figure 28: Survey parameters: confidence level and design prevalence.

Then, the user will be asked to provide the total number of inspection units in the green survey band.

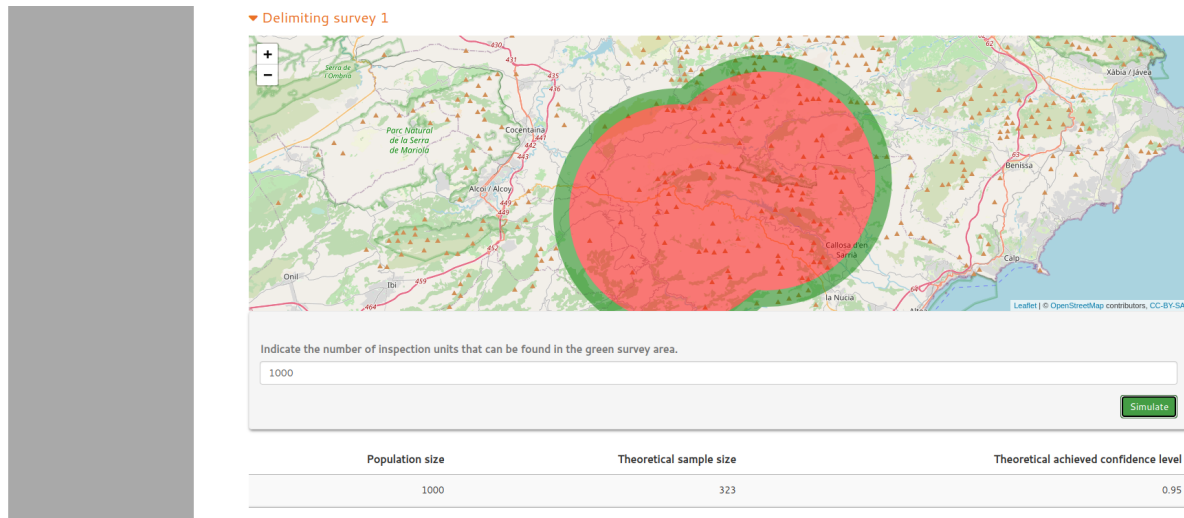


Figure 29: The first PIZ with survey band.

Next, the user will have to go through the allocation step. The user has two options.

Option 1, the user will have to list every possible inspection unit in the survey band. The tool will then pick the samples to be inspected at random. Then the user can do the field work and provide the results back to the tool.

Option 2, the user can directly provide the samples taken from the field with their location and if the sample was either positive (1) or negative (0).

In case in the first survey band, a positive finding has been reported. Then, the PIZ will increase. In case a negative finding has been reported, the PIZ will decrease.

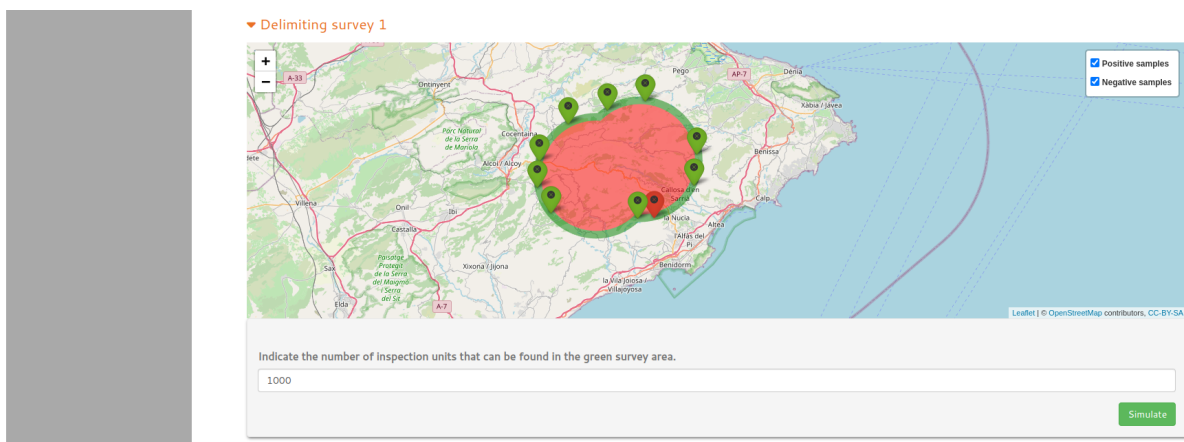


Figure 30: An example of reported findings by the user.

In the example above, one positive finding has been reported (red marker). All the green markers are negative findings reported by the user. In this case, the PIZ will increase until all findings in the survey band are negative. Thus, the user will have to potentially repeat this process a few times to get to the desired infested zone.

The design phase consists of a second part, the summary. There, the user will see the final map of the infested zone and the user will also be able to download the information needed for the buffer zone survey. Keep in mind that if the user wants to start a buffer zone survey the session has to be restarted by pressing the 'house'-button in the top right corner.

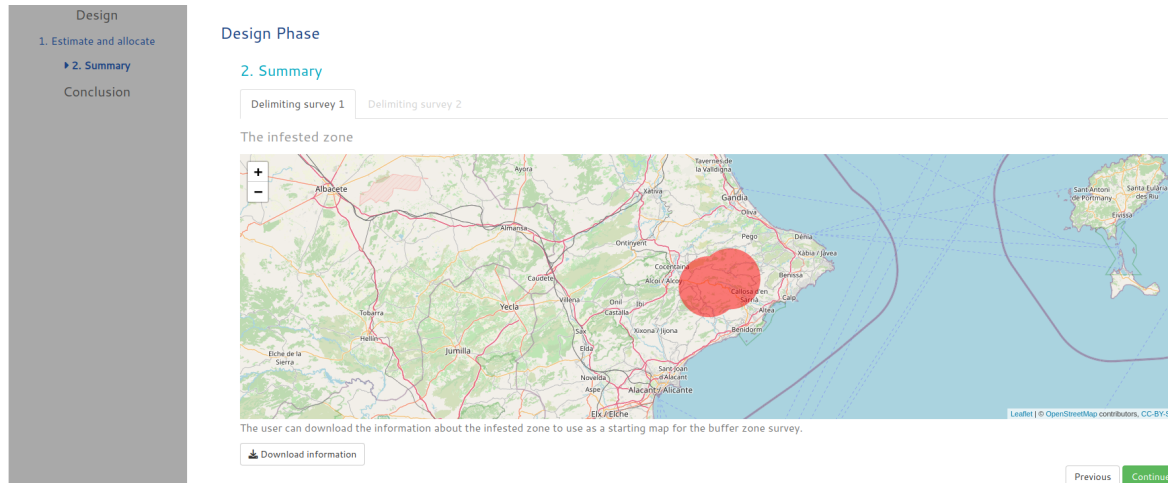


Figure 31: The summary of the delimiting survey.

3.4 Conclusion

In the conclusion a report in MS Word can be downloaded.

4 BUFFER ZONE SURVEY

The buffer zone survey, like the detection survey, is also divided into an initiation phase, a preparation phase, a design phase, an implementation phase and a conclusion.

4.1 Initiation Phase

The initiation phase is similar to the initiation phase of the delimiting survey. Keep in mind that all questions need to be answered, unless they are indicated as optional.

4.2 Preparation phase

The preparation phase is divided into three subsections: 1. Define infested zone, 2. Target population and 3. Inspection units.

In the example below, the infested zone from the delimiting survey has been uploaded.

1. Define infested zone

2. Approach

3. Inspection units

Design

Implementation

Conclusion

Preparation phase

1. Define infested zone

What is the upper range of the yearly spread capacity of the pest? [m]

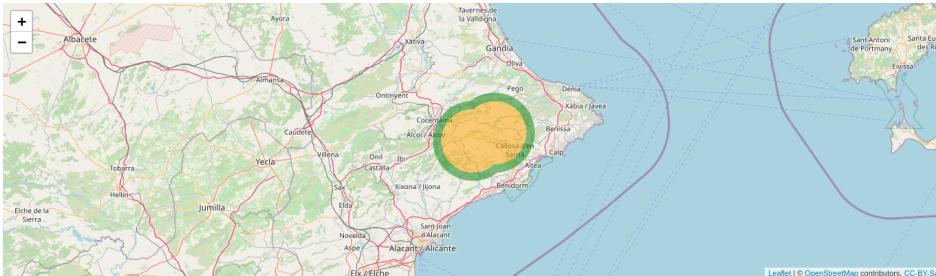
Do you have a file with information about the infested zone from this tool?

☒ Yes ☐ No

Upload the json file

Browse... start_bufferZoneSurvey_2023-09-11.json

Upload complete



Previous Next

Figure 32: Define the infested zone based on the delimiting survey.

The user is also able to construct the buffer zone completely from scratch. As can be seen in the Figure below.

2. Target population

3. Inspection units

Design

Implementation

Conclusion

Preparation phase

1. Define infested zone

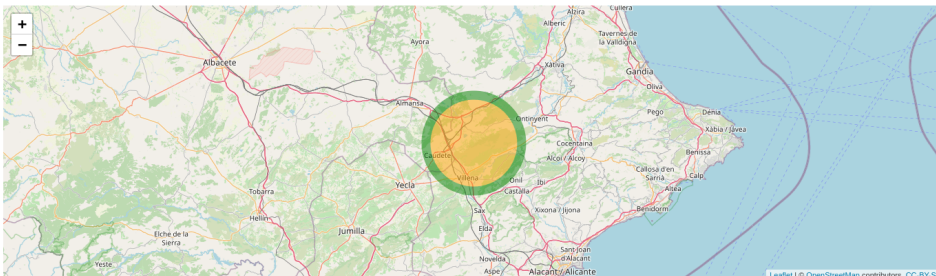
What is the upper range of the yearly spread capacity of the pest? [m]

Do you have a file with information about the infested zone from this tool?

☐ Yes ☒ No

How many centers does the infested zone contain?

	Label	Latitude	Longitude	Radius [m]
1	Finding 1	38.744200	-0.842000	15000.00



Previous Next

Figure 33: Define the infested zone from scratch.

From the subsection '2. Target population', the buffer zone is the same as the detection survey and the user can refer to those parts in the manual if needed.